COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure & Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	Jul 10, 2026	**DATE SUBMITTED:**	Jul 12, 2026
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ACTIVITY NO. 1.1 **[Seatwork - Algorithm, Flowchart, Pseudocode]**

1. PROCEDURE OUTPUT

What procedures did you perform?

TASK 1:
- Performed creating a program where the user input and computes its score on their three quizzes and also after getting the sum, it will divide it to 3 to get the average using Do/While and Switch Case.

TASK 2: Performed to create a program (cashiering system) where the user input the name of three item, input his/her balance, and also the price of the three items after putting the three items and its prices when the program starts and also add a feature where if the user have insufficient balance, it will ask the user if it want to add a balance so the transaction will be successful, but if not, the program will restart. Using Do/While and Switch Case.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

(The flowchart, pseudocode and algorithm and output of it will be separated of this file because of the format of this file when I insert it, it becomes messy and hard to fix for me)

1. Compute the total and average of 3 quizzes then display the total and average.

Code Analysis:

I declared a variable named int choice and a bool that is equal to true and I created a Do/While on it so the program will keep running unless the user's choice will be "2".After the user input its choice I created a Switch Case, where inside it was the choice so each case will depend on the user's input and in case 1 I declared a 3 integer with a zero value for the user to input its score on his/her quiz and I created three user input for quiz 1 to quiz 3 and the user will input his score on each of it. After that, I created an integer variable name "sum" where I just added / (+) operator on each score on quiz based on the user's input and after that I created a float with a variable name "average" where the "sum" variable was inside it and also the total score of the three quiz and I divided it to 3.0 to get the average on it and it's the end of the case 1. In case 2 when the user chooses it, it will stop the program because the bool named "run" is true and I made it equal to false (run = false) so the Do/While loop will stop inside the Switch Case. In default, it will say "Error, Please choose again." and will return on the program where the user asks to input its choice.

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Code Analysis:

First, I declared a variable string named item1-item3 for the name of the items, float price1-price3, for the price of each item, choice for the user's input choice where he/she wants to proceed, bool run equals to true for the Do/While loop, character confirmation if the user wants to add a balance, float sum_price for the total price of each item, and for the change is the total balance left after the transaction of your current balance subtracting for the total sum for the three items. It will ask the user first to add his/her balance and I created a Do/While Loop and make a choice of create a transaction and let the user input its choice and i put a Switch Case, and inside of it was the choice the user will input for each condition inside based on what the user will input and inside the case 1, where the user let put its three items and the price each of it and i initialize the price 1-price3 inside the sum_price and add an (+) operator to get the total and subtract the balance to the sum_price and created an if-else statement where the balance is greater or equal to the sum_price and used (-=) on it so the current balance will be shown after the transaction and not the old one and will display the total, current balance and the change will receive of the user after the transaction, while the else statement will display the user have not enough balance to pay based on the total price of each item and ask the user if if it want to add a balance and if the user input "y" on the character variable and i created an if statement again that it will add a balance on it to its insufficient balance, it will display the balance that he add and the program will restart. While in default, it will say its invalid and the program will restart from the start.

Conclusion:

In this seatwork, I learned and grasped what I learned back then in C++ last year. In this activity, I also learned how to use pseudocode and flowchart based on the algorithm, learned how to use multiple statements and conditions again based on the given problem even though it took me days to grasp it again, it's a big help for me to understand C++ basic programming again with this activity and ready for the next seatwork or hands-on activity.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini (AI Search from Chrome)

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Code Assistance

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

COMPUTER ENGINEERING

QUEZON CITY

Extent of Use: Asking AI to inspect my code and pinpointing where areas need attention like adding, removing, missing on my code, or where did i go wrong (ex. missing curly brackets, missing variable on specific area/condition, forgot to add a specific variable/curly bracket on that area. Also saying “don’t correct”,”don’t give suggestions”, and ”don’t give the final output of the correct code” because without pointing that out, AI will automatically give the final output and I said that to my prompt so the AI will avoid giving it to me.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.’s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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